Predicting Chronic Illness in Aging Populations Using Physiological Markers

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Abstract

Objective: Examine whether machine learning (ML) can predict chronic illness in older adults using wearable and clinical data.

Data: 21 Fitbit + electronic health record (EHR) elements --> comprise measures of physical activity, cardiovascular health, and sleep for almost 2,000 participants in the All of Us Research Program (aged 60+).

XGBoost, Random Forest, and Logistic Regression were the models used. Class imbalance was addressed through validation using cross-validation and upsampling.

Key Findings: → XGBoost outperformed other models in osteoporosis (AUC = 0.802, F1 = 0.742); → The models struggled in dementia and heart failure since there were few positive findings.

In conclusion, ML models hold great promise for passive, early risk identification. especially favorable for musculoskeletal disorders such as osteoporosis

Introduction

Chronic illness is the leading cause of disability in older adults:

- 92% of Americans aged 60+ live with at least one chronic condition
- 80% live with two or more[^1]

For immediate handling and to avoid loss of independence, early detection is essential.

Wearable devices like Fitbit now offer passive, real-time monitoring of:

- Physical activity
- Sleep behaviour
- Cardiovascular signals

Before clinical symptoms appear, machine learning (ML) can reveal subtle, predicted patterns in certain physiological measures.

This study uses data from the All of Us (AOU) Research Program to predict:

- Osteoporosis (musculoskeletal)
- Dementia (neurocognitive)
- Heart failure (cardiovascular)

The objective is to evaluate the predictive value of wearable and clinical data for the categorization of chronic illnesses in older adults using machine learning models.

Methodology

Data Source

- All of Us (AOU) Research Program – Controlled Tier Dataset (v8)
- Participants: ~2,000 individuals aged 60+
- Data types: Fitbit wearable summaries + Electronic Health Records (EHR)

Feature Engineering (21 Total Features)

1. Physical Activity
   - Daily steps, calories burned, BMR, floors climbed
   - Sedentary vs. active minutes, elevation gain
2. Cardiovascular Metrics
   - Min/Max heart rate
   - Minutes in heart rate zones
   - Systolic & diastolic blood pressure
3. Sleep Behaviour
   - REM, light, deep, and wake durations
   - Total sleep time
4. Clinical Indicators
   - Body Mass Index (BMI), body weight

Modelling Strategy

- Algorithms:
  - Logistic Regression (LM)
  - Random Forest (RF)
  - XGBoost (XGB)
- Training:
  - 75/25 train-test split
  - 5-fold cross-validation
  - Upsampling applied to training data
- Evaluation Metrics:
  - Accuracy
  - F1-score (balances precision & recall)
  - AUC (discriminative ability)

Technical Challenges

- **Severe class imbalance**
  → Dementia and heart failure had very few positive cases
  → Led to **undefined F1 scores** and unreliable generalization
- **Data sparsity**
  → Many participants lacked full wearable data
  → Final sample dropped from ~60,000 to ~2,000
- **Train/test imbalance**
  → Some test sets had most of the positive cases
  → Artificially inflated AUC despite poor recall
- **Workaround strategies**
  - Upsampling during training
  - Stratified sampling
  - Median imputation for missing values

Results

Model Performance by Condition

Condition	Best Model	Accuracy	F1 Score	AUC
Osteoporosis	XGBoost	**0.745**	**0.742**	**0.802**
Dementia	XGBoost	0.986	—	0.579
Heart Failure	XGBoost	0.897	—	0.678

The model with the best prediction performance for osteoporosis Dependable classification is demonstrated by high AUC and F1-score => Best features: BMI, BMR, and calories burnt

Dementia:
→ High accuracy, but no specified F1-score → Shows inability to recall positive dementia cases
→ Most predictive: REM sleep, light sleep, and total bedtime

Heart Failure:
→ Significant class imbalance
→ Undefined F1-scores
→ Moderate AUC values; the best predictors are the length of the HR zone, REM sleep, and active minutes.

Top Features of Model Performance (XGBoost)

Condition	Top Features
Osteoporosis	BMR, daily calories burned, BMI
Dementia	REM sleep, light sleep, time in bed
Heart Failure	Active minutes, REM sleep, HR zones

Osteoporosis

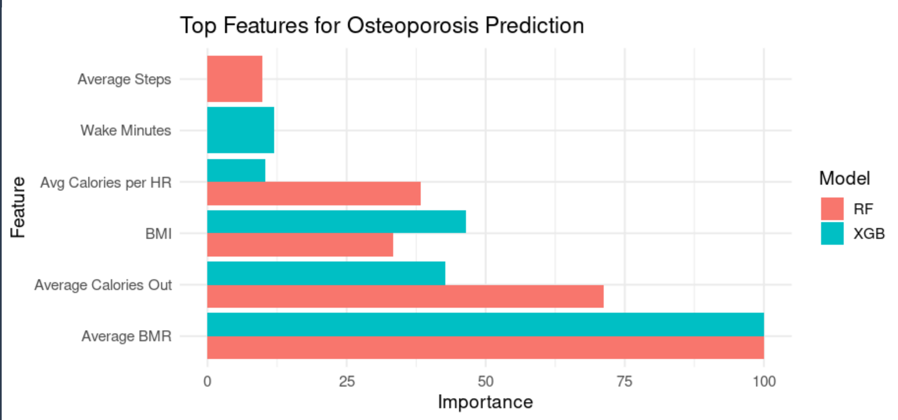
- Basal Metabolic Rate (BMR): Strongest negative association with risk
- Calories Burned: Higher daily expenditure linked to lower risk
- BMI: Elevated BMI was mildly protective, reflecting lean mass contribution

Dementia

- REM Sleep Duration: Lower REM time associated with higher risk
- Light Sleep & Time in Bed: Disruptions in sleep architecture may signal early cognitive decline
- Heart Rate Variability: Showed emerging relevance across models

Heart Failure

- Fairly Active Minutes: Lower daily movement linked to condition presence
- REM Sleep: Reduced REM patterns correlated with cardiovascular strain
- Time in Heart Rate Zones: Less time in moderate zones suggested lower fitness



Random Forest identified minimum heart rate, step count, and floors climbed as the strongest predictors of osteoporosis risk, suggesting that daily activity and cardiovascular health play key roles in musculoskeletal outcomes.

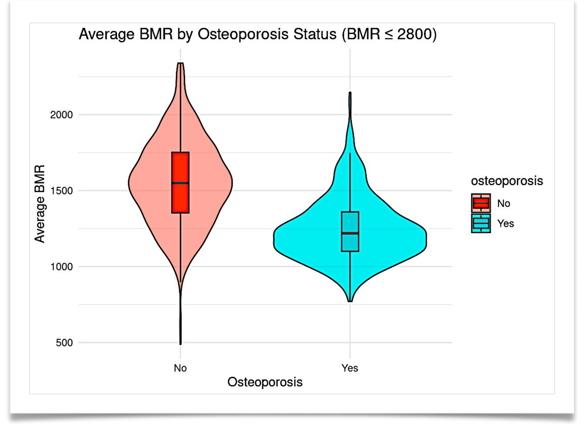
Conclusion & Future Insights

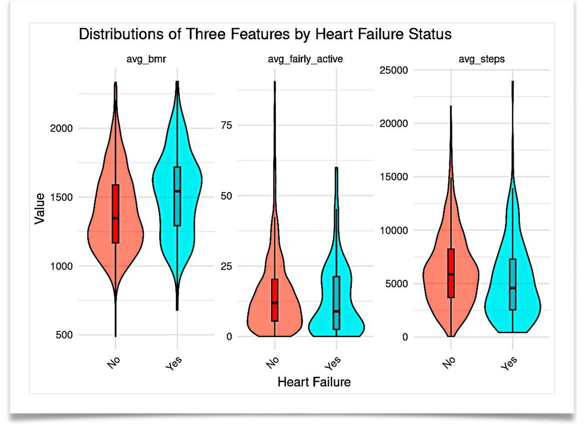
Key Takeaways
Using wearable technology and medical information, machine learning can predict an individual's risk of developing a chronic condition.

The best prediction accuracy was found for osteoporosis: → XGBoost AUC = 0.802, F1 = 0.742 → Basal Metabolic Rate (BMR) was the most prominent feature.

Class imbalance limited models of heart failure and dementia.

Physiological Feature Trends
Osteoporosis: Reduced BMR in afflicted individuals validates established metabolic connections.
Heart Failure: Reduced activity is a distinctive early indication for heart failure, as evidenced by lower fairly active minutes and step count in positive patients.





The promise for early identification utilizing wearable data is reinforced by this distribution, which shows a strong physiological distinction across groups.

Future Directions

- bigger, more balanced datasets
- Engineering features specific to a disease
- For uncommon conditions, synthetic minority sampling (like SMOTE)

References & Acknowledgements

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